

Education

- 2021–2026 : **Ph.D. in Mathematics**, *Vanderbilt University*, United States.
Advisor: Akram Aldroubi
Research interests: Applied and computational harmonic analysis, sampling theory, signal processing, and inverse problems.
- 2017–2020 : **M.Sc. in Pure Mathematics**, *Beijing Normal University*, China.
Advisor: Xiaojing Xu
Research interests: PDEs in fluid dynamics
- 2013–2017 : **B.Sc. in Mathematics and Applied Mathematics**, *Beijing Normal University*, China.

Publications

Preprints

- 2026 Le Gong. Identification and recovery of linear dynamics from partial trajectories. 2026. Submitted.
- 2026 Akram Aldroubi, Rocío Díaz Martín, Le Gong, Javad Mashreghi, and Ivan Medri. Characterization of frames for source recovery from dynamical samples. 2026. Submitted.

Published Articles

- 2026 Le Gong and Longxiu Huang. Randomized space–time sampling for affine graph dynamical systems. *IEEE Transactions on Information Theory*, volume 72, pages 112–126, 2026.
- 2025 Akram Aldroubi, Rocío Díaz Martín, Le Gong, Javad Mashreghi, and Ivan Medri. Frames for source recovery in dynamical systems. In *2025 International Conference on Sampling Theory and Applications (SampTA)*, pages 1–3, 2025.
- 2025 Akram Aldroubi, Le Gong, Ilya Krishtal, Brendan Miller, and Sumati Thareja. Reconstruction algorithms for source term recovery from dynamical samples in catalyst models. *Journal of Fourier Analysis and Applications*, volume 31, page 53, 2025.
- 2023 Le Gong, Shiyong Li, Naqib Sad Pathan, Mohammad Shifat-E-Rabbi, Gustavo K. Rohde, Abu Hasnat Mohammad Rubaiyat, and Sumati Thareja. The radon signed cumulative distribution transform and its applications in classification of signed images. *Poincare Journal of Analysis & Applications*, volume 10, pages 89–114, 2023.
- 2023 Akram Aldroubi, Le Gong, and Ilya Krishtal. Recovery of rapidly decaying source terms from dynamical samples in evolution equations. *Sampling Theory, Signal Processing, and Data Analysis*, volume 21, page 15, 2023.

Talks

- Invited Talk **AMS 2026 Spring Southeastern Sectional Meeting**, *Savannah, GA, US*, March 28–29, 2026.
- Invited Talk **Joint Mathematics Meetings**, *Washington, DC, US*, January 4–7, 2026.
- Invited Talk **Mathematical Congress of the Americas 2025**, *Miami, FL, US*, July 21–25, 2025.
- Contributed Talk **Southeastern Atlantic Section of SIAM Section Meeting 2025**, *University of Tennessee, Knoxville, TN, US*, March 21–23, 2025.

- Invited Talk **Analysis seminar 2025**, *Vanderbilt University, Nashville, TN, US*, February 3, 2025.
- Poster **Prairie Analysis Seminar 2024**, *University of Kansas, Lawrence, KS, US*, October 25-26, 2024.
- Invited Talk **2024 International Workshop on Frames, Operators and Related Topics**, *Chern Institute of Mathematics, Nankai University, Tianjin, China*, July 16-18, 2024.
- Poster **KUMUNU-ISU 2024 PDE, Dynamical Systems and Applications**, *University of Kansas, Lawrence, KS, US*, April 6-7, 2024.
- Invited Talk **American Mathematical Society (AMS) 2024 Spring Southeastern Sectional Meeting**, *Florida State University, Tallahassee, FL, US*, March 23–24, 2024.
- Poster **Fall Fourier Talks 2023**, *Norbert Wiener Center, University of Maryland, College Park, MD, US*, October 26-27, 2023.

Awards

- 2026 Bjarni Jonsson Prize for Research, Vanderbilt University.
- 2026 Travel Grant, Joint Mathematics Meetings, AMS.
- 2025 Travel Grant, Mathematical Congress of the Americas (MCA), AMS/NSF.

Academic Service

- Reviewer São Paulo Journal of Mathematical Sciences
- Reviewer Sampling Theory, Signal Processing, and Data Analysis

Workshops/Summer Schools

- October 2024 Frame Theory Days 2024, Saint Louis University, St. Louis, MO, US.
- June 2023 SLMath Summer Graduate School on Machine Learning, University of California, San Diego, CA, US.

Computer skills

- Programming Python, Matlab
- Languages

Teaching Assistantship

- Fall 2025 : **Math 1301: Accelerated Single-Variable Calculus II**, Vanderbilt University.
- Spring 2025 : **Math 2400: Differential Equations with Linear Algebra**, Vanderbilt University.
- Fall, 2024 : **Math 1200: Single-Variable Calculus I**, Vanderbilt University.
- Spring 2024 : **Math 2400: Differential Equations with Linear Algebra**, Vanderbilt University.
- Fall 2023 : **Math 1201: Single-Variable Calculus II**, Vanderbilt University.
- Spring 2023 : **Math 1301: Accelerated Single-Variable Calculus II**, Vanderbilt University.
- Fall 2022 : **Math 1301: Accelerated Single-Variable Calculus II**, Vanderbilt University.
- Spring 2022 : **Led study hall and worked at the teaching center**, Vanderbilt University.
- Fall 2021 : **Led study hall and worked at the teaching center**, Vanderbilt University.